Modeling electrical losses in CsPbI3 solar cells: stacked idealizations and a routine-quality envelope at 18.0%

Hussain Touhid Siddiquee[*,a] and Md. Abdul Malek Fahim[a]

[a]Department of EEE, Leading University, Sylhet-3112, Bangladesh

[*]Corresponding author. E-mail: touhidsiddiqueeraj@gmail.com

Article type: Regular Paper, for consideration in *Solar Energy* (Elsevier).

Abstract. Stacked idealisations inflate SCAPS-1D CsPbI3 projections. On one fully worked CsPbI3/CBTS stack, joint optimisation of thickness, doping, defects, gap, and both transport-layer affinities gives a permissive-box ceiling of **24.95%**. Sampling a frozen routine-quality prior volume (96-point Latin hypercube over eight factors; ranges and measured anchors in Table 1 of the manuscript) gives a median of **18.0%** with a 5–95% prior-predictive envelope of **15.6–20.0%**. This is a sensitivity envelope under stated priors, not a confidence interval on real devices. The 2024–2026 experimental records fall above that envelope. The ceiling record comparison is density-of-states conditional: with capped density of states, literature radiative recombination ($C_r = 10^{-10}$ cm^3/s) lowers the ceiling to 21.86%, below the 22.02% certified record; with effective-mass physical density of states, the same radiative term gives 23.46% ($N_v = 4.1 \times 10^{18}$ cm^{-3}) and 23.04% ($N_v = 8.9 \times 10^{18}$ cm^{-3}), above it. Interface sensitivity is regime dependent, bulk defects dominate at routine defect density, and series resistance up to 5 Ωcm^2 closes roughly half of the six-point fill-factor gap against measured devices, with the remainder attributed to nonradiative interface recombination (ideality above unity).

Keywords: CsPbI3; perovskite solar cell; SCAPS-1D; CBTS; reproducibility; device simulation

Author contributions (CRediT). Hussain Touhid Siddiquee: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, validation, and writing—original draft, review and editing. Md. Abdul Malek Fahim: software, validation, and writing—original draft.

Data availability. The simulation archive is based on repository revision `6e991c1`. Definition files, sweep receipts, raw curves, scripts, verification receipts, and typeset sources are in the public repository at `https://github.com/touhidsiddiqueeraj-bit/Modeling-electricModeling-electrical-losses`